

SIMATS ENGINEERING
ASSESSMENT TOOL 1: Data Interpretation

COURSE NAME: COMPUTER VISION

COURSE CODE: ITA0515

Slot B

COURSE OUTCOME COVERED

CO4: Compare object and pattern recognition techniques for interpreting visual data with spatial and motion characteristics. (BTL 5)

Assessment Tool Weightage: 40%

QUESTIONS:5

Total Marks 50

Duration :60 Minutes

Annexure A

Data Interpretation

Code of Conduct:

I S.Jeswanth Kumar (192321056) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: S.Jeswanth Kumar

Rubrics for Evaluation

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Marks(100)
Understanding of Data	25%	Accurately interprets all data patterns, trends, and relationships	Interprets data correctly with minor gaps	Partial understanding; misses key trends	Misinterprets or fails to understand data	
Analysis	25%	Provides deep analysis with strong logical reasoning and justification	Provides reasonable analysis with some justification	Basic analysis with weak reasoning	No meaningful analysis provided	
Application of Concepts	20%	Effectively applies relevant concepts/tools to interpret data accurately	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts to data	
Conclusion	20%	Draws clear, meaningful conclusions with strong insights and implications	Provides valid conclusions with moderate insights	Conclusions are vague or weak	No clear or valid conclusions	
Clarity	10%	Presents interpretation clearly using proper structure, visuals, and terminology	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand	

Read the given data carefully and compare the techniques or results based on multiple factors such as accuracy, robustness, and computational constraints. Evaluate and justify your decision with appropriate reasoning considering the application context.

1. Structure from Motion Consistency

Two datasets produce:

- Dataset A: accurate depth but requires high-quality images
- Dataset B: inconsistent depth but works on low-quality images

Compare the datasets and evaluate which setup is more practical for real-world deployment. Justify your reasoning.

2. Stereo Vision vs Motion Estimation

Depth estimation results:

- Stereo Vision: accurate but requires dual cameras
- Motion Estimation: less accurate but works with single camera

Compare both approaches and evaluate which is preferable under hardware constraints. Justify your choice.

3. Model Generalization vs Overfitting

Performance:

Model Training Testing

A	98%	70%
B	90%	88%

Compare the models and evaluate which is more reliable for deployment. Justify based on generalization and risk factors.

4. YOLO Performance across Scales

Detection results:

- Large objects: 95%
- Medium objects: 85%
- Small objects: 60%

Compare performance across scales and evaluate whether YOLO alone is sufficient for multi-scale detection tasks. Justify improvements.

5. Motion Model Selection for Animation

Two models:

- Model A: smooth but physically inaccurate
- Model B: realistic but computationally intensive

Compare the models and evaluate which should be used for animation systems under different constraints (real-time vs cinematic). Justify your decision.

ANSWERS

Q1. Structure from Motion (SfM) Consistency

Comparison

Factor	Dataset A	Dataset B
Depth accuracy	High / accurate	Low / inconsistent
Input requirement	High-quality images needed	Works on low-quality images
Robustness to real-world capt	uLroew (fails under poor conditions)	Higher (tolerant of noise/blur)
Computational/data cost	Higher (needs careful acquisition)	Lower (flexible capture)

Evaluation and Justification

Dataset A gives more accurate depth reconstruction but only under controlled, high-quality imaging conditions, which real-world deployments (handheld cameras, drones, outdoor lighting, motion blur) cannot always guarantee. Dataset B sacrifices some accuracy but continues to function when the input quality degrades, which is exactly the situation most field deployments face. Since Structure from Motion is typically used in uncontrolled environments (mapping, robotics, AR on consumer devices), robustness to input variability is usually more valuable than peak accuracy under ideal conditions.

Conclusion: Dataset B's setup is more practical for real-world deployment because graceful degradation under poor image quality outweighs the higher but conditional accuracy of Dataset A. In applications where precision is critical (e.g., survey-grade 3D reconstruction), Dataset A would still be preferred, so the choice is context-dependent but favors B for general deployment.

Q2. Stereo Vision vs Motion Estimation

Comparison

Factor	Stereo Vision	Motion Estimation
Depth accuracy	High	Lower
Hardware requirement	Dual (calibrated) camera rig	Single camera
Cost & form factor	Higher cost, larger footprint	Lower cost, compact
Calibration/maintenance	Needs precise stereo calibration	Simpler setup

Evaluation and Justification

Stereo vision produces more accurate depth because it directly triangulates from two synchronized viewpoints, but it requires a dual-camera rig with careful calibration, adding cost, size, and power consumption. Motion estimation (structure from motion / optical-flow-based depth) uses a single moving camera, making it far cheaper and easier to integrate into constrained platforms such as small drones, mobile phones, or embedded IoT devices, at the cost of reduced accuracy and sensitivity to motion quality.

Conclusion: Under hardware constraints (limited space, weight, power, or budget), motion estimation is preferable because a single-camera solution is deployable where a stereo rig physically or economically cannot be. Where hardware is not a constraint and accuracy is critical (e.g., robotic grasping, autonomous vehicles), stereo vision remains the better choice.

Q3. Model Generalization vs Overfitting

Comparison

Model	Training Accuracy	Testing Accuracy	Train-Test Gap
A	98%	70%	28% (large)
B	90%	88%	2% (small)

Evaluation and Justification

Model A has very high training accuracy (98%) but its testing accuracy drops sharply to 70%, a 28-point gap that is a classic signature of overfitting: the model has memorized training-set patterns and noise rather than learning generalizable features, so its real-world/unseen-data performance is unreliable.

Model B has a smaller gap between training (90%) and testing (88%) accuracy, indicating it has learned features that transfer well to new data, even though its peak training accuracy is lower.

Conclusion: Model B is more reliable for deployment. Deployment performance is governed by behaviour on unseen data, and a small, consistent train-test gap indicates predictable, trustworthy performance in production, whereas Model A's large gap poses a high risk of failure on real-world inputs despite its impressive training score. Regularization, data augmentation, or early stopping could be applied to Model A to close the gap before it is considered deployment-ready.

Q4. YOLO Performance across Scales

Comparison

Object Scale	Detection Accuracy
Large objects	95%
Medium objects	85%
Small objects	60%

Evaluation and Justification

Accuracy falls steadily as object size decreases (95% * 85% * 60%). This is a well-known limitation of single-shot detectors like YOLO: small objects occupy very few pixels after repeated downsampling in the backbone/feature-pyramid, so their feature representation becomes weak, coarse grid cells cause localization error, and small objects are more easily suppressed or confused with background during non-max suppression.

Conclusion: YOLO alone is not sufficient for reliable multi-scale detection, particularly for small objects. Improvements that address this include: using a finer feature-pyramid / multi-scale detection head (e.g., FPN/PANet, as in later YOLO versions), increasing input resolution for small-object-heavy scenes, tiling/cropping large images (slicing-aided inference), adding anchor sizes tuned to small objects, and applying data augmentation (mosaic, small-object copy-paste) to increase small-object representation during training.

Q5. Motion Model Selection for Animation (10 Marks)

Comparison

Factor	Model A	Model B
Motion quality	Smooth	Physically realistic

Physical accuracy	Physically inaccurate	High
Computational cost	Low	High (computationally intensive)
Best suited for	Real-time applications	Offline / cinematic rendering

Evaluation and Justification

Model A prioritizes visual smoothness over physical correctness and is computationally cheap, making it well suited to real-time systems (games, interactive simulations, VR) where consistent frame rate and responsiveness matter more than perfect physical fidelity, and minor physical inaccuracies are rarely noticed by users during interaction. Model B produces physically realistic motion by solving more accurate (and expensive) physics/dynamics, which suits offline cinematic production where render time is not constrained by a real-time frame budget and visual/physical realism is the priority for the audience.

Conclusion: The choice is context-dependent rather than absolute: Model A should be used for real-time animation systems (games, live simulations) where speed and interactivity are critical, while Model B should be used for cinematic/offline animation where computation time is available and realism is paramount. A hybrid approach (simplified physics for real-time preview, high-fidelity physics for final render) is also a common practical compromise.